

Pact: Autonomous Multi-Agent Arbitration

A Verifiable Protocol for Resolving Contractual Disputes

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Abstract. We propose a protocol and system architecture for autonomous resolution of contractual disputes using adversarial multi-agent deliberation. The core innovation is a hierarchical agent architecture that simulates judicial proceedings while providing verifiable procedural constraints. By grounding legal reasoning in an immutable constitutional layer, specifically the UNIDROIT Principles of International Commercial Contracts, the system bounds legal interpretation and prevents ungrounded citation or hallucinated doctrine. We introduce *Adversarial Symmetry Verification* to detect asymmetric outcomes under party relabeling and *Precedent-Aware Reasoning* to improve temporal consistency across decisions. The protocol commits to evidence and process via cryptographic hashes and supports deterministic replay through locked context and decision seeds, producing a standardized arbitration award alongside a compact verification bundle suitable for audit and enforcement workflows. The system is designed to reduce the marginal cost of dispute resolution dramatically. This extends procedural safeguards to disputes that are currently uneconomic to pursue and contributes infrastructure toward closing the global access-to-justice gap.

1. Introduction

The modern legal system faces a scaling constraint: the supply of qualified adjudication is limited, while the demand for resolution spans billions of routine disputes that are economically irrational to litigate or arbitrate using human labor. Traditional dispute resolution mechanisms, litigation and arbitration, are reliable, but they are time-intensive, expensive, and geographically constrained. In many commercial settings, the cost of resolution can exceed the value at stake, turning contract enforcement into negotiation under asymmetric leverage rather than adjudication under procedural safeguards.

This paper presents Pact, a system and protocol for autonomous resolution of contractual disputes. Pact does not claim to replace human judgment in high-stakes or novel matters. Instead, it defines a procedurally constrained arbitration substrate for disputes that are currently underserved by institutional pathways. The design objective is to preserve the legitimacy drivers of adjudication, symmetry, bounded discretion, and auditable reasoning, while reducing marginal costs through automation.

Pact is presented as a protocol rather than a single “prompt to answer” model. The central unit is a committed record and an executable procedure that transforms inputs into a court-facing award and verification artifacts. The system is engineered for adversarial settings and treats integrity, replayability, and enforcement-oriented output formats as first-class requirements.

2. The Problem with Current Approaches

2.1 Rule-Based Systems

Early online dispute resolution systems attempted to encode legal reasoning into fixed rules. This approach does not scale because contract language and fact patterns form a combinatorial space that is effectively unbounded, because interpretive reasoning depends on context and ambiguity, and because maintaining a rulebase under evolving doctrine imposes a continuous expert burden.

2.2 Single-Model AI Approaches

Applying a single large language model to dispute resolution fails on structural grounds. Unconstrained generation permits hallucinated citations and fabricated legal principles, repeated runs can yield inconsistent outcomes without replay guarantees, decisions lack adversarial testing, and biases embedded in training data can propagate into outcomes unless explicitly detected and corrected by protocol-level mechanisms.

2.3 Human Arbitration

Traditional arbitration is procedurally robust but cannot scale to the long tail of disputes. Costs for arbitrators, administration, and legal representation create high barriers to entry; even expedited processes require weeks to months; and cross-border disputes face choice-of-law and enforcement complexity.

3. System Architecture

Pact is implemented as a hierarchical multi-agent system designed to separate concerns, enforce procedural symmetry, and produce auditable artifacts. The architecture includes orchestration, trial execution, core agent roles, and post-processing.

3.1 Architectural Layers

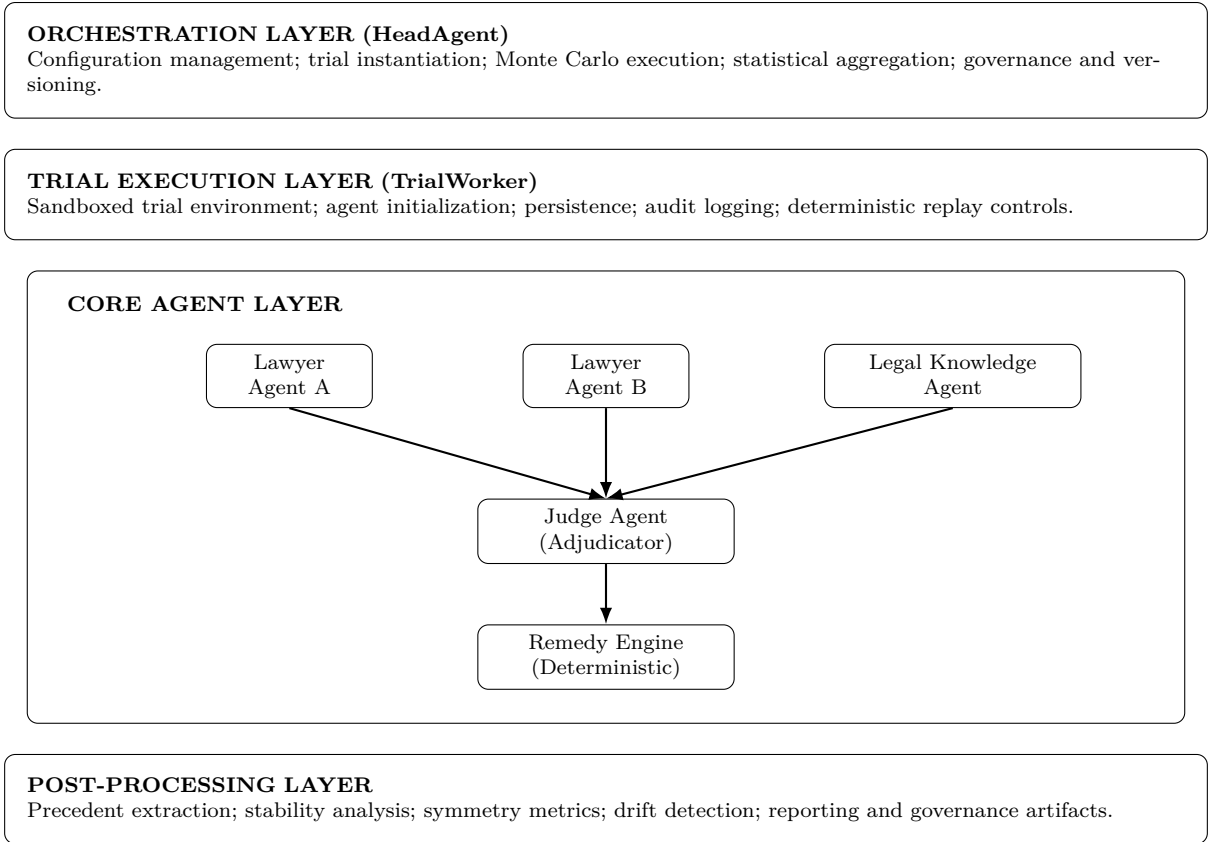


Figure 1: Layered architecture and core agent roles.

3.2 Agent Roles and Responsibilities

Evidence ingestion transforms raw inputs into a normalized case record while preserving original materials. In a typical deployment, an Evidence Agent parses the contract and artifacts, extracts obligations and timelines, identifies asserted breaches, and produces structured summaries that support bounded retrieval. The raw artifacts remain authoritative; summaries exist to reduce context length while preserving traceable references back to original items.

Each party is represented by a Lawyer Agent tasked with constructing arguments favorable to its assigned party under explicit strategy profiles. Strategy profiles constrain permissible argument classes, evidentiary emphasis, and response behavior. This creates substantive adversarial diversity rather than merely stylistic variation.

The Legal Knowledge Agent serves as the constitutional constraint layer. It answers strictly within a defined legal corpus and refuses out-of-scope queries, bounding doctrine and preventing hallucinated legal sources.

The Judge Agent controls the trial, frames issues, tracks burdens, queries party and legal agents, and produces structured decision artifacts. The Remedy Engine computes remedies deterministically from those artifacts, separating liability from numerical remedy calculation.

3.2.1 Evidence Request Agent

After analyzing the contract, the briefs, and the current record, an Evidence Request Agent conducts structured evidence elicitation with each party. The agent generates a case-specific evidence checklist derived from disputed clauses, required performance elements, timelines, and claimed damages. It then asks targeted questions in the style of counsel, requesting missing documents, receipts, invoices, correspondence, delivery confirmations, specifications, or other exhibits likely to resolve factual gaps. Each request is recorded, responses are hashed and added to the committed record, and failures to produce requested categories of evidence are tracked for subsequent burden analysis and credibility assessment.

4. Protocol Objects, Commitments, and Case Record

Pact defines protocol primitives that anchor integrity and auditability. These primitives bind the award to committed inputs and committed process, enabling verification that the stated procedure was applied to the stated record.

4.1 Evidence Artifacts and Merkle Commitment

An evidence artifact is an immutable byte sequence with content hash. Let $H(\cdot)$ denote a cryptographic hash function. For artifact E_i , the identifier is $\text{eid}_i = H(E_i)$. The set of evidence identifiers is committed by a Merkle root $R_{\text{evidence}} = \text{MerkleRoot}(\text{eid}_1, \dots, \text{eid}_n)$ under a canonical ordering rule.

4.2 Process Commitment

Before substantive adjudication begins, Pact commits to process configuration. The committed process hash H_{proc} binds procedure specification, artifact schemas, prompts, model/version identifiers, retrieval rules, and safety boundaries. The committed legal corpus hash H_{corpus} binds the constitutional text and version.

4.3 Case Header

Field	Meaning
<code>case_id</code>	Unique identifier for the case instance
<code>t₀</code>	Commitment timestamp for case start
<code>H_{contract}</code>	Hash of the governing contract canonical form
<code>R_{evidence}</code>	Merkle root committing to evidence artifact hashes
<code>H_{corpus}</code>	Hash of constitutional legal corpus and excerpts
<code>H_{proc}</code>	Hash of procedure specification and schemas
<code>H_{strategy}</code>	Hash of party strategy profiles
<code>seed</code>	Decision seed for deterministic replay
<code>H_{award}</code>	Hash of final award and artifact set

Table 1: Case Header schema.

4.4 Evidence Commitment Diagram

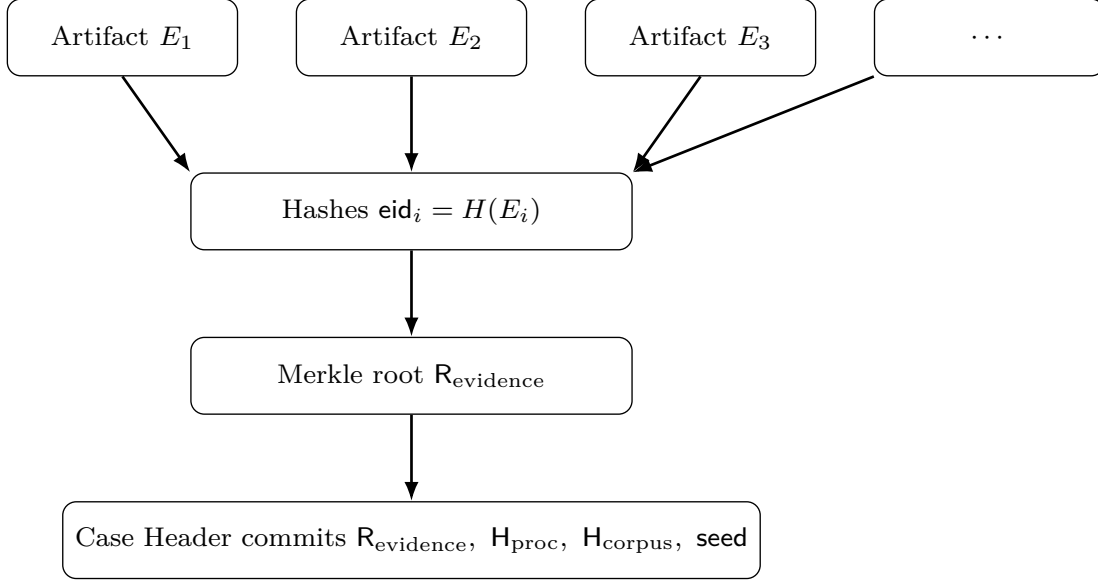


Figure 2: Evidence commitment and anchoring of process and corpus.

4.5 Proof of Fair Process Bundle

Pact outputs a verification bundle that binds the award to committed inputs and committed process. Let the bundle be

$$\Pi = (H_{\text{header}}, H_{\text{proc}}, H_{\text{corpus}}, H_{\text{strategy}}, R_{\text{evidence}}, \text{seed}, H_{\text{award}}).$$

Optional deployments may publish commitments to an append-only transparency log, enabling third parties to verify that decisions were produced under the committed process.

5. The Trial Protocol

5.1 Turn-Bounded Deliberation

The Judge Agent conducts a turn-bounded loop in which it selects actions to query party agents or the legal knowledge agent and ultimately to decide. Bounded rounds guarantee termination and limit procedural abuse.

5.2 Decision Frame, Burden Ledger, and Judicial Artifacts

Pact replaces stored chain-of-thought with structured judicial artifacts that are stable, comparable, and enforcement-oriented. The Judge emits numbered Findings of Fact with evidence citations, Findings of Law mapped to constitutional provisions, an Application Map linking specific facts to legal elements, and a conclusion. A Decision Frame schema captures adjudication commitments

about contract clarity, performance state, breach classification, evidentiary strength by side, activated legal provisions, and remedy eligibility.

The protocol makes burden of proof a tracked state. Each disputed issue is recorded in a burden ledger with the burdened party and a status that progresses through met, not met, or disputed. Remedies requiring breach, causation, or quantified loss are gated on burden status.

5.3 Evidence Scoring and Conflict Resolution

Each evidence artifact is scored for relevance, authenticity signals, specificity, and corroboration within the record. When conflicting evidence bears on a material fact, the Judge emits a conflict resolution entry citing artifacts and rationale grounded in scoring and corroboration structure.

5.4 Protocol Dataflow Diagram

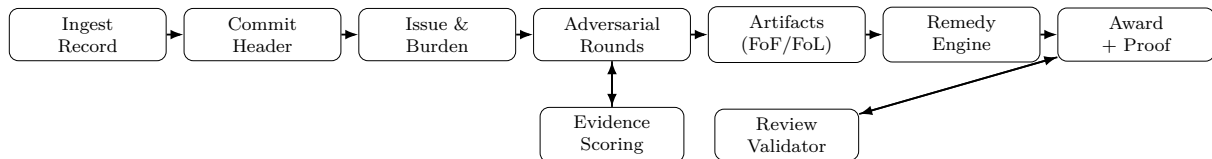


Figure 3: End-to-end dataflow from committed record to award and verification.

5.5 State Machine Diagram

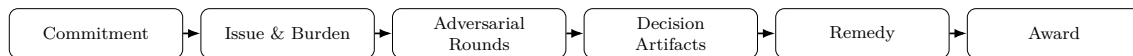


Figure 4: Turn-bounded arbitration state machine.

6. Constitutional Grounding and Norm Hierarchy

Large language models can generate plausible legal text without guarantees of correctness. In legal contexts this manifests as fabricated statutes, misapplied doctrine, or hallucinated precedents. Pact addresses this through constitutional grounding: the Legal Knowledge Agent is constrained to an immutable corpus, specifically the UNIDROIT Principles of International Commercial Contracts, and must refuse questions outside scope.

This creates a closed-world assumption for law. The system can only apply rules that exist in its constitutional layer, and the source of those rules is verifiable through the committed corpus hash H_{corpus} . The Judge cites constitutional provisions when producing Findings of Law and maps their elements to Findings of Fact via the Application Map.

Pact enforces hierarchical norm resolution. Constitutional principles constrain interpretation and override party argumentation; contract terms define obligations within the constitutional frame; party claims are evaluated within contract and constitution. When conflicts arise, the decision artifacts explicitly resolve the conflict according to this hierarchy.

7. Statistical Assurance and Bias Mitigation

Pact treats fairness and stability as properties to be measured and governed, not merely asserted. The orchestration layer supports controlled re-execution, population-level measurement, and drift detection using protocol-native artifacts.

7.1 Adversarial Symmetry Verification

A core bias diagnostic is Adversarial Symmetry Verification. The same dispute is executed under original party assignments and then re-executed under party relabeling or inversion. Artifact-level deltas are computed and compared under a configured tolerance. Large divergences trigger review. Across a population of cases, systematic divergence patterns indicate representational bias or procedural asymmetry and motivate correction.

7.2 Monte Carlo Stability and Sensitivity Analysis

Pact supports Monte Carlo execution as a stability probe. The HeadAgent runs many trials of the same case under controlled parameter perturbations such as judge philosophy variants, lawyer strategy variants, stochastic sampling settings, retrieval ordering perturbations, and summarization seeds. The output is a distribution over artifact sets rather than a single outcome. Stability can be measured as agreement rates on liability findings, remedy bands, and key fact determinations. Sensitivity analysis identifies which parameters materially influence outcomes and which are benign, enabling governance decisions about defaults and acceptable variance.

7.3 Multi-Run Aggregation and Panel Adjudication

An optional execution mode uses a judge panel rather than a single judge. Multiple Judge Agents execute the same committed record independently under distinct judicial philosophies. A meta-aggregation step produces a majority outcome and preserves dissent artifacts. This resembles appellate dynamics in miniature and reduces dependence on a single model instantiation. The same mechanism can be used to compute confidence measures derived from inter-judge agreement.

7.4 Identifier Masking and Invariance Testing

To reduce identity-linked bias, deployments can apply an anonymization pipeline that replaces names and identifying details with neutral placeholders while preserving the evidentiary structure. Invariance testing can then measure whether outcomes change materially when identifiers are masked or permuted, using artifact deltas as the comparison surface.

8. Precedent-Aware Reasoning

A legitimate dispute resolution system must be consistent over time. Pact introduces Precedent-Aware Reasoning as an extension that creates a living body of machine-readable precedent. Decisions are embedded into a vector space along with structured metadata about dispute type, clauses, factual patterns, and outcomes. When a new dispute arrives, similar decisions are retrieved and

included as context for adjudication. The Judge explicitly states whether it follows or distinguishes retrieved precedents, producing an auditable trail for consistency.

Human override can be integrated as high-authority precedent. When a human arbitrator reviews and corrects a decision, the correction is captured as a precedent artifact with elevated weight in future retrieval and consistency checks.

9. Verifiable Process Integrity and Deterministic Replay

Parties must trust not only the outcome but the process. Pact commits to process configuration before adjudication begins by hashing prompts, schemas, model identifiers, corpus versions, and procedural rules into H_{proc} and H_{corpus} . This enables verification that a given award corresponds to the committed procedure and corpus.

Deterministic replay is supported through locked context and a decision seed in the Case Header. Given the committed record, the committed process, and the seed, the system reproduces identical artifacts. Replay enables audit, debugging, regression testing, and resolution of disputes about process compliance.

10. Explainability Architecture

Opaque decisions lack legitimacy. Pact produces explanations at multiple abstraction levels derived from the same underlying artifacts. Technical explanations include complete trial logs, committed configuration identifiers, and artifact-level validation results. Legal explanations include applicable constitutional provisions, the elements satisfied, the evidence cited, and the argument resolution rationale. Plain-language explanations summarize the dispute, identify decisive factors, and describe what would have changed the outcome.

Counterfactual explanations are supported through controlled re-execution with modified inputs. Counterfactual runs are separate from the committed case and are labeled as hypothetical; they do not modify the original decision record.

11. Privacy and Security Architecture

Contract disputes involve sensitive information. Deployments can apply an anonymization pipeline that scrubs personal identifiers and optionally company names, producing blind evaluation while maintaining reversible mappings held securely. Trials execute in isolated sandboxes with strict data separation, explicit retention policies, and encryption at rest, including customer-managed keys where required.

Prompt injection is a primary adversarial vector. Pact mitigates injection by enforcing strict delimiter boundaries between procedure and party submissions, sanitizing inputs, applying anomaly detection, and validating outputs against expected artifact schemas. These measures reduce the probability of unsafe instruction following and make deviations detectable through artifact validation.

12. Correctness Properties and Validators

Pact benefits from protocol-native correctness properties that can be verified offline against stored artifacts. Such properties include the requirement that every Finding of Fact cites committed evidence identifiers; that every Finding of Law cites constitutional provisions and identifies their elements; that the Application Map links each legal conclusion to specific facts; that no remedy is computed unless the relevant burdens are satisfied; that conflict resolutions are present when material evidence conflicts exist; and that the award hash binds to the committed record and process hashes in the Case Header.

Validator	Artifact-level condition
Evidence binding	Every cited exhibit resolves to a member proof under R_{evidence}
Law binding	Every legal rule cited resolves to H_{corpus} and named provisions
Fact support	Each Finding of Fact cites exhibits sufficient to justify the statement
Burden gating	Remedies requiring breach or loss are blocked unless burdens are satisfied
Conflict accounting	Material conflicts produce explicit conflict-resolution artifacts
Award binding	The award and artifacts hash to H_{award} recorded in the Case Header

Table 2: Illustrative correctness validators.

13. Economic Model

Traditional arbitration cost decomposes into arbitrator time, administration, and representation. Pact shifts cost toward computation and storage. The marginal cost of a trial is dominated by model inference calls and bounded orchestration overhead, and can be reduced further through batching, caching, and retrieval minimization. This enables a new category of enforceable dispute resolution for low-value disputes that are currently abandoned as uneconomic.

To discourage frivolous filings, deployments can incorporate filing deposits refundable upon substantial success, disclosure windows, cost-shifting policies for bad-faith behavior, and reputation mechanisms for repeat participants. These policies are deployment parameters rather than protocol primitives.

14. Legal Framework Integration

Pact is designed for contractual integration through an arbitration clause that specifies the protocol as the arbitral forum and binds parties to final outcomes. The award output is structured as an arbitration instrument with clear operative clauses suitable for submission to courts where enforcement is sought.

Because Pact is constitutionally grounded in UNIDROIT, it can operate in a jurisdiction-agnostic mode appropriate for international commercial contracts. For blockchain-based agreements, Pact can optionally integrate with escrow or smart contract triggers, allowing awards to initiate automated execution under agreed constraints, while preserving separation between decision and enforcement authority.

15. Limitations and Future Work

Pact is optimized for documentary commercial disputes. It cannot verify physical evidence, compel testimony, or substitute for live cross-examination where credibility is central. Its legal scope is bounded by the constitutional corpus. Multilingual support requires careful evaluation to avoid semantic drift. Precedent-aware reasoning requires governance to prevent accumulation of inconsistent or low-quality precedent artifacts.

Future work includes multimodal evidence ingestion with provenance controls, witness interaction modes that preserve procedural integrity, internal review by panel configurations, pluggable constitutional corpora for other legal traditions, and mediation-style early intervention before full adjudication.

16. Conclusion

Pact presents a protocol and architecture for autonomous contract dispute resolution that addresses the limitations of rule-based systems, single-model approaches, and unscalable human arbitration. Its core contributions are hierarchical multi-agent adjudication, constitutional grounding in UNIDROIT, adversarial symmetry verification, Monte Carlo stability and sensitivity measurement, optional panel adjudication, precedent-aware reasoning, cryptographic process commitments, deterministic replay, and correctness validators. The objective is not to automate judgment everywhere just yet, but to extend structured dispute resolution to cases that existing systems cannot economically serve.

References

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- [5] Bai, Y., et al. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*. 2022.
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A. UNIDROIT Principles Coverage

The UNIDROIT Principles provide broad coverage of commercial contract law, including general provisions, formation and authority, validity, interpretation, contractual content and third party rights, performance, non-performance, set-off, assignment and transfer, limitation periods, and plurality of obligors and obligees.

B. Illustrative Arbitration Clause

Any dispute, controversy, or claim arising out of or relating to this Agreement shall be finally resolved by arbitration conducted pursuant to the Pact protocol in effect at the time of dispute initiation. The parties expressly consent to resolution by an automated arbitration system and to a final and binding arbitration award. Judgment upon the award may be entered in any court of competent jurisdiction.

C. System Parameters

Parameter	Value	Rationale
max_turns	50	Enables thorough examination while bounding runtime and cost
judge_temperature	0.3	Lower variance to improve stability of decision artifacts
lawyer_temperature	0.7	Higher variance to broaden adversarial argument search
confidence_threshold	0.90	Below this threshold triggers escalation or review
symmetry_tolerance	0.10	Maximum acceptable artifact divergence in ASV

Table 3: Default configuration parameters.

D. Illustrative Proof of Fair Process Verification

An auditor verifies that the award and decision artifacts hash to H_{award} and that all referenced evidence artifacts are members of the Merkle tree committed by R_{evidence} . The auditor verifies that the procedure and corpus versions used match H_{proc} and H_{corpus} recorded in the Case Header and that the decision seed matches seed for replayability. Optional deployments may add signatures or transparency-log inclusion proofs.

E. System Architecture

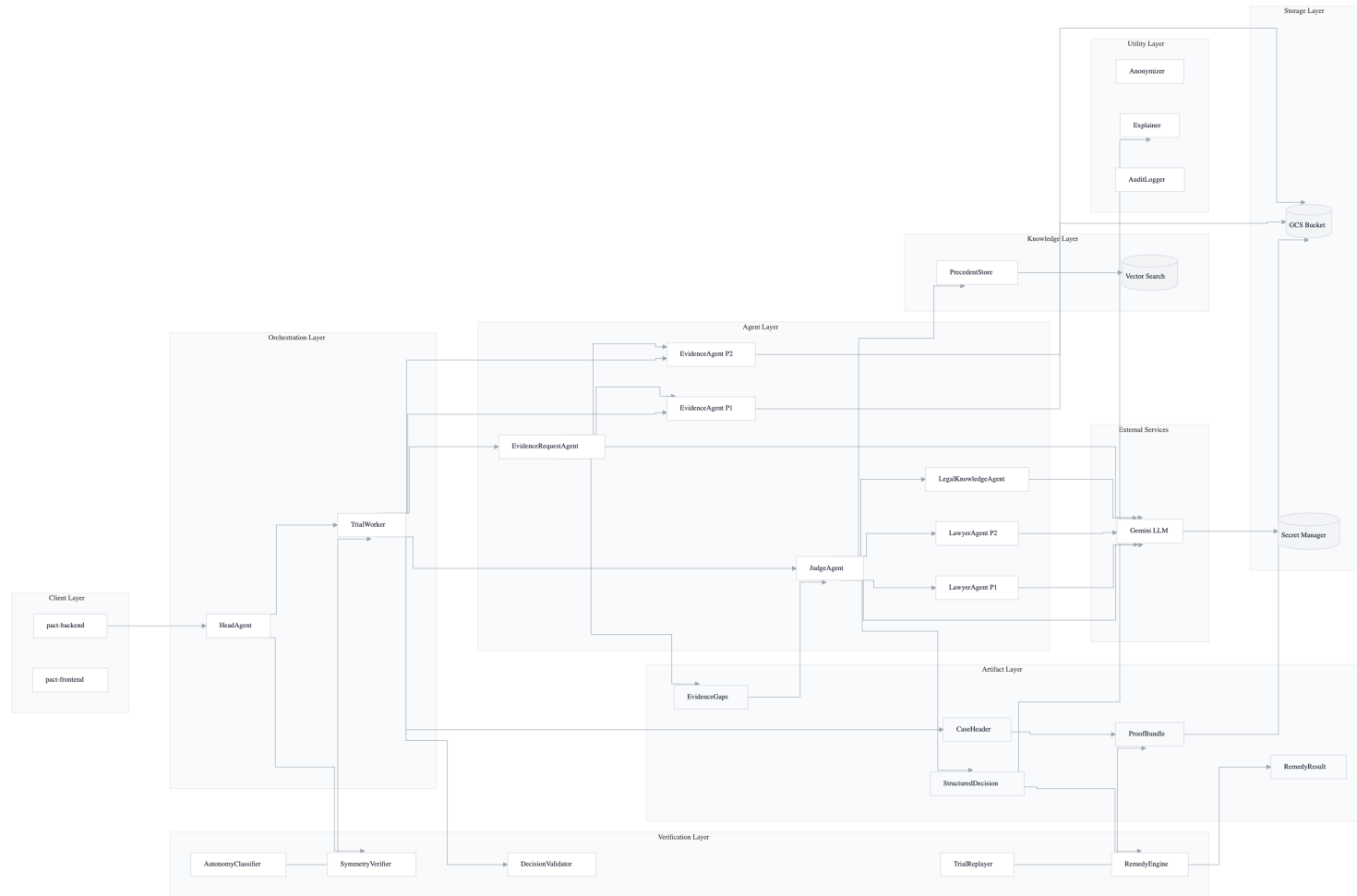


Figure 5: System Architecture